Chapter 118

Stereo Nodes

This chapter details the Stereo nodes available in Fusion. Stereoscopic nodes are available only in Fusion Studio and DaVinci Resolve Studio.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

Contents

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Anaglyph [Ana]

The Anaglyph node

NOTE: The Anaglyph node is available only in Fusion Studio and DaVinci Resolve Studio.

Anaglyph Node Introduction

The Anaglyph node is used to create stereoscopic images by combining separate left eye and right eye images. It is most commonly used at the end of a stereoscopic workflow to display or deliver the final result.

Inputs

The three inputs on the Anaglyph node are the left eye input, right eye input, and effect mask.

Left Eye Input: The orange input is used to connect the 2D image representing the left eye in the stereo comp.

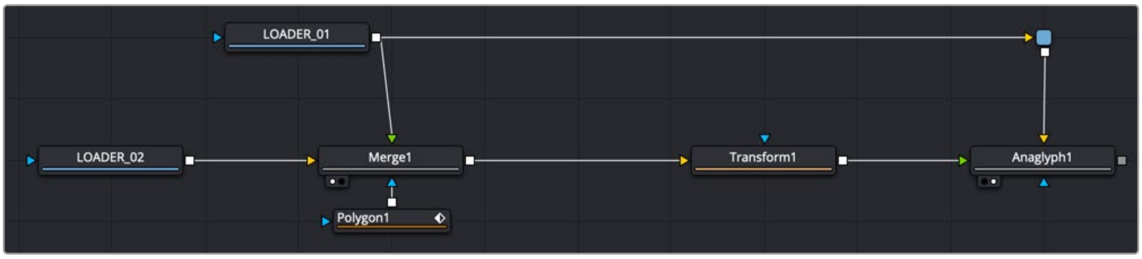
Right Eye Input: The green input is used to connect the 2D image representing the right eye in the stereo comp.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the stereoscopic creation to only those pixels within the mask.

Basic Node Setup

The Anaglyph node is usually placed at the end of a stereoscopic node tree to display the final result.

When using separate images for the left and right eye, the left eye image is connected to the orange input, and the right eye image is connected to the green input of the node. When using either horizontally or vertically stacked images containing both left eye and right eye information, these only connect to the orange input.



An Anaglyph node using separate left and right eye inputs

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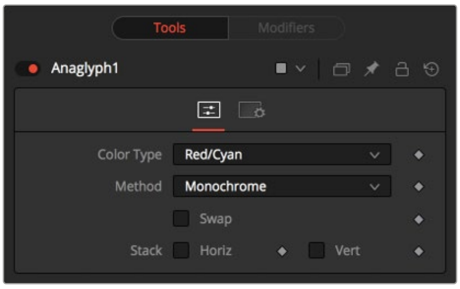
Inspector



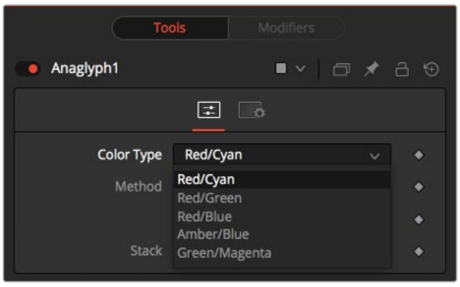
The Anaglyph Controls tab

Controls Tab

Using the parameters in the Controls tab, the separate images are combined to create a stereoscopic output.

Color Type Menu

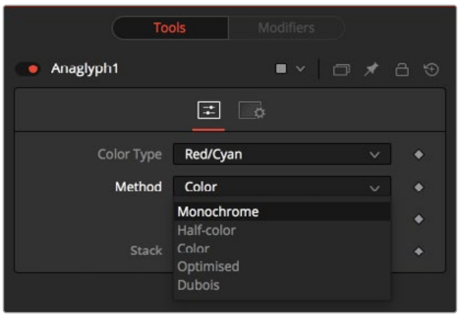
The Color Type menu allows you to choose between different color encodings to fit your preferred display device. To match your stereo glasses, you can choose between Red/Cyan, Red/Green, Red/Blue, Amber/Blue, and Green/Magenta encoding; Red/Cyan is the most commonly used.



The Anaglyph Color Type menu

Method

In addition to the color used for encoding the image, you can also choose five different methods from the Method menu: Monochrome, Half-color, Color, Optimized, and Dubois. These methods are described below.



The Anaglyph Method menu

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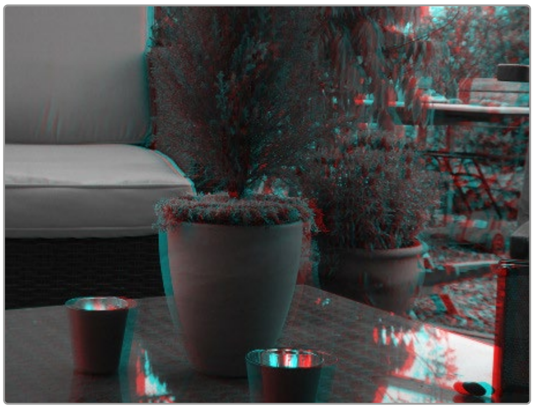
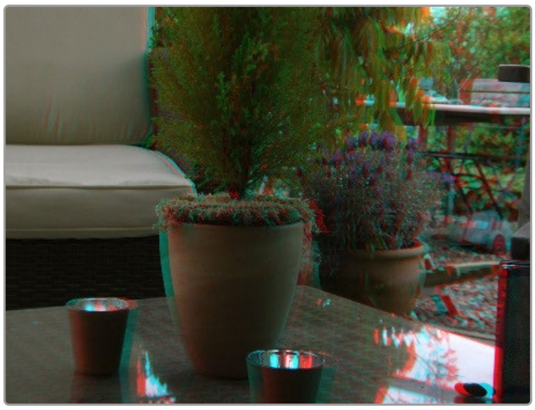
- **Monochrome:** Assuming you are using a Red/Cyan Color Type, the left eye contains the luminance of the left image and is placed in the output of the red channel. The right eye contains the luminance of the right image and is placed in the output green and blue channels.
- **Half-Color:** Assuming you are using a Red/Cyan Color Type, the left eye contains the luminance of the left image and is placed in the output of the red channel. The right eye contains the color channels from the right image that match the glasses' color for that eye.



Monochrome



Half-Color

- **Color:** The left eye contains the color channels from the left image that match the glasses' color for that eye. The right eye contains the color channels from the right image that match the glasses' color for that eye.
- **Optimized:** Used with red/cyan glasses, for example, the resulting brightness of what shows through the left eye is substantially less than the brightness of the right eye. Using typical ITU-R 601 ratios for luminance as a guide, the red eye would give 0.299 brightness, while the cyan eye would give 0.587+0.114=0.701 brightness—over twice as bright. The difference in brightness between the eyes can produce what are referred to as retinal rivalry or binocular rivalry, which can destroy the stereo effect. The Optimized method generates the right eye in the same fashion as the Color method. The left eye also uses the green and blue channels but in combination with increased brightness that reduces retinal rivalry. Since it uses the same two channels from each of the source images, it doesn't reproduce the remaining one. For example, 1.05× the green and 0.45× the blue channels of the left image is placed in the red output channel, and the green and blue channels of the right image are placed in the output green and blue channels. Red from both the left and right images is not used.

Color

Optimized

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- **Dubois:** Images with fairly saturated colors can produce retinal rivalry with the Half-color, Color, and Optimized methods because the color is visible in only one eye. For example, with red/cyan glasses, a saturated green object looks black in the red eye, and green in the cyan eye. The Dubois method uses the spectral characteristics of (specifically) red/cyan glasses and CRT (Trinitron) phosphors to produce a better anaglyph and in the end, tends to reduce retinal rivalry caused by such color differences in each eye. It also tends to reduce ghosting produced when one eye 'leaks' into the other eye. The particular calculated matrix in Fusion is designed for red/cyan glasses and isn't available for other glasses types. Since it is also derived from CRT color primaries, it may not give the best results with a common LCD (though it'll still likely produce less retinal rivalry and ghosting than the other methods).

Dubois

Swap Eyes

Allows you to swap the left and right eye inputs easily.

Horiz Stack

Takes an image that contains both left and right eye information stacked horizontally. These images are often referred to as "crosseyed" or "straight stereo" images. You only need to connect that one image to the orange input of the node. It then creates an image half the width of the original input, using the left half of the original image for the left eye and the right half of the original image for the right eye. Color encoding takes place using the specified color type and method.

Vert Stack

Takes an image that contains both left and right eye information stacked vertically. You only need to connect that one image to the orange input of the node. It then creates an image half the height of the original input, using the bottom half of the original image for the left eye and the top half of the original image for the right eye. Color encoding takes place using the specified color type and method.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Combiner [Com]

The Combiner node

NOTE: The Combiner node is available only in Fusion Studio and DaVinci Resolve Studio.

Combiner Node Introduction

The Combiner node takes two stereoscopic inputs and creates so-called "stacked images" with the left and right eye, either side by side or on top of each other. Stereoscopic nodes are available only in Fusion Studio and DaVinci Resolve Studio.

Inputs

The two inputs on the Combiner node are used to connect the two images that get combined in a stacked stereo image.

Image 1 Input: The orange input is used to connect the 2D image representing the left eye in the stereo comp.

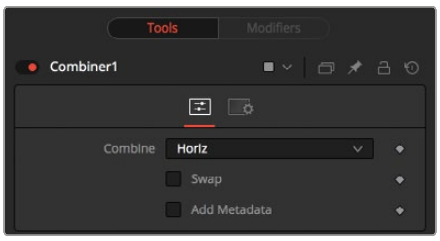
Image 2 Input: The green input is used to connect the 2D image representing the right eye in the stereo comp.

Basic Node Setup

Below, a left eye image and right eye image are connected to the Combiner node to create a single stacked stereo image. It can be more efficient to render out the stacked stereo images as EXR files than to generate the disparity on-the-fly.

Left and right eye images are connected into a Combiner node to generate a stacked stereo image.

Inspector



The Combiner Controls tab

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Controls Tab

To stack the images, the left eye image is connected to the orange input, and the right eye image is connected to the green input of the node.

Combine

The Combine menu provides three options for how the two images are made into a stacked stereo image.

- **None:** No operation will take place. The output image is identical to the left eye input.
- **Horiz:** Both images will be stacked horizontally, or side-by-side, with the image connected to the left eye input on the left. This will result in an output image double the width of the input image.
- **Vert:** Both images will be stacked vertically, or on top of each other, with the image connected to the left eye input on the bottom. This will result in an output image double the height of the input image.
- **Layers:** This exposes layer name fields, allowing you to rename the layers.

Swap Eyes

Allows you to easily swap the left and right eye input.

Add Metadata

Metadata is carried along with the images and can be added to the existing metadata using this checkbox. To view Metadata, use the viewer's SubView menu set to Metadata.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Disparity [Dis]

The Disparity node

NOTE: The Disparity node is available only in Fusion Studio and DaVinci Resolve Studio.

Disparity Node Introduction

Disparity generates the left/right shift between the frames in a stereo pair. It also generates the vertical disparity between the left/right images, which is usually a lot smaller than the horizontal disparity and ideally should be 0 to minimize viewing discomfort. When viewing the output of the Disparity node in the views, the human eye can distinguish quality/detail in the Disparity map better by looking at either the grayscale X disparity or Y disparity rather than looking at the combined XY disparity as a Red/Green color image.

The generated disparity is stored in the output image's Disparity aux channel, where the left image contains the left > right disparity, and the right image contains the right > left disparity. Because disparity works based on matching regions in the left eye to regions in the right eye by comparing colors and gradients of colors, colors in the two eyes must be as similar as possible. Thus, it is a good idea to color correct ahead of time. It is also a good idea to crop away any black borders around the frames, as this confuses the disparity tracking (and also causes problems if you are using the Color Corrector's histogram match ability to do the color matching).

In Stack mode, left and right outputs deliver the same image. If the left and right images have a global vertical offset larger than a few pixels, it can help the disparity tracking algorithm if you vertically align features in the left/right eyes ahead of time using a Transform node. Small details tend to get lost in the tracking process when you have a large vertical offset between left/right eyes.

Consider using a SmoothMotion node to smooth your disparity channel. This can help reduce time-dependent flickering when warping an eye. Also, think about whether you want to remove lens distortion before computing disparity. If you do not, your Disparity map becomes a combined Disparity and Lens Distortion map. This can have advantages and disadvantages.

One disadvantage is that if you then do a vertical alignment, you are also removing lens distortion effects. When trying to reduce the computation time, start first with adjusting the Proxy and Number of Iterations sliders.

The Disparity node does not support RoI or DoD.

Inputs

The two inputs on the Disparity node are used to connect the left and right images.

Left Input: The orange input is used to connect either the left eye image or the stacked image.

Right Input: The green input is used to connect the right eye image. This input is available only when the Stack Mode menu is set to Separate.

Outputs

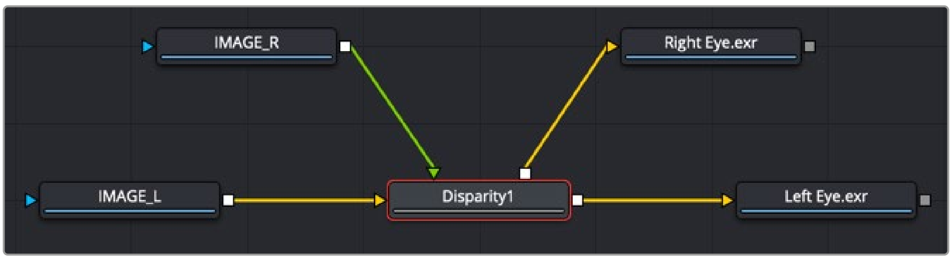
Unlike most nodes in Fusion, Disparity has two outputs for the left and right eye.

Left Output: This holds the left eye image with a new disparity channel, or a Stacked Mode image with a new disparity channel.

Right Output: This holds the right eye image with a new disparity channel. This output is visible only if Stack Mode is set to Separate.

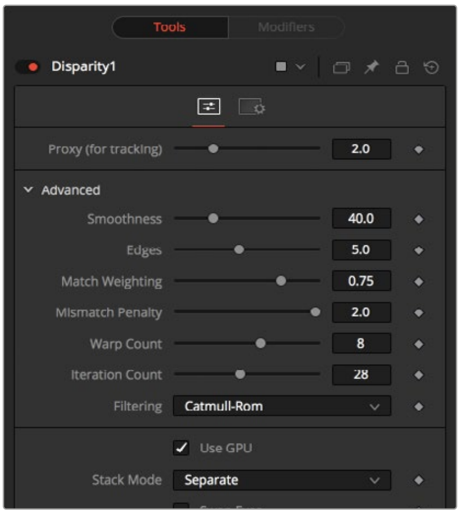
Basic Node Setup

Below, a left eye image and right eye image are connected to the Disparity node. The Disparity node then outputs each eye to Saver nodes. It can be more efficient to render out the stereo images as EXR files than to generate the disparity on-the-fly.



Left and right eye images are connected into a Disparity node
to generate and render out a stereo image

Inspector



The Disparity Controls tab

Proxy (for Tracking)

The input images are resized down by the proxy scale, tracked to produce the disparity, and then the resulting disparities are scaled back up. This option is purely to speed up the calculation of the disparity, which can be slow. The computational time is roughly proportional to the number of pixels in the image. This means a proxy scale of 2 gives a 4x speedup, and a proxy scale of 3 gives a 9x speedup. In general, 1:1 proxy will give the most detailed flow, but keep in mind that this is highly dependent on the amount of noise and film grain. If noise is present in large quantities, it can completely obliterate any gains moving from 2:1 to 1:1 proxy. In some situations, it can even make things worse. You can think of the Proxy setting as acting as a simplistic low-pass filter for removing noise/grain.

Advanced

The Advanced settings section has parameter controls to tune the Disparity map calculations. The default settings have been chosen to be the best default values from experimentation with many different shots and should serve as a good standard. In most cases, tweaking of the Advanced settings is not needed.

Smoothness

This controls the smoothness of the disparity. Higher values help deal with noise, while lower values bring out more detail.

Edges

This slider is another control for smoothness but applies it based on the color channel. It tends to have the effect of determining how edges in the disparity follow edges in the color images. When it is set to a lower value, the disparity becomes smoother and tends to overshoot edges. When it is set to a higher value, edges in the disparity align more tightly with the edges in the color images, and details from the color channels start to slip into the disparity, which is not usually desirable.

As a rough guideline, if you are using the disparity to produce a Z channel for post effects like depth of field, experiment with higher values, but if you are using the disparity to do interpolation, you might want to keep the values lower.

In general, if the Edges slider is set is too high, there can be problems with streaked out edges when the disparity is used for interpolation.

Match Weight

This controls how matching is done between neighboring pixels in the left image and neighboring pixels in the right image. When a lower value is used, large structural color features are matched. When higher values are used, small sharp variations in the color are matched. Typically, a good value for this slider is in the [0.7, 0.9] range. Setting this option higher tends to improve the matching results in the presence of differences due to smoothly varying shadows or local lighting variations between the left and right images. You should still color match the initial images so they are as similar as possible; this option tends to help with local variations (e.g., lighting differences due to light passing through a mirror rig).

Mismatch Penalty

This controls how the penalty for mismatched regions grows as they become more dissimilar. The slider provides a choice between a balance of Quadratic (lower values) and Linear (higher values) penalties. Lower value Quadratic settings strongly penalize large dissimilarities, while higher value Linear settings are more robust to dissimilar matches. Moving this slider toward lower tends to give a disparity with more small random variations in it, while higher values produce smoother, more visually pleasing results.

Warp Count

Turning down the Warp Count makes the disparity computations faster. In particular, the computational time depends linearly upon this option. To understand what this option does, you need to understand that the Disparity algorithm progressively warps the left image until it matches with the right image. After some point, convergence is reached, and additional warps are just a waste of computational time. The default value in Fusion is set high enough that convergence should always be reached. You can tweak this value to speed up the computations, but it is good to watch how the disparity is degrading in quality at the same time.

Iteration Count

Turning down the Iteration Count makes the disparity computations faster. In particular, the computational time depends linearly upon this option. Just like adjusting Warp Count, at some point adjusting this option higher will yield diminishing returns and will not produce significantly better results. By default, this value is set to something that should converge for all possible shots and can be tweaked lower fairly often without reducing the disparity's quality.

Filtering

This menu determines the filtering operations used during flow generation. Catmull-Rom filtering will produce better results, but at the same time, it increases the computation time steeply.

Stack Mode

This menu determines how the input images are stacked.

When set to Separate, the Right Input and Output will appear, and separate left and right images must be connected.

Swap Eyes

Enabling this checkbox causes the left and right images to swap.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Disparity To Z [D2Z]

The DisparityToZ node

NOTE: The Disparity to Z node is available only in Fusion Studio and DaVinci Resolve Studio.

Disparity to Z Node Introduction

Disparity To Z takes a 3D camera and an image containing a disparity channel as inputs, and outputs the same image but with a newly computed Z channel.

Optionally, this node can output Z into the RGB channels. Ideally, either a stereo Camera 3D or a tracked stereo camera is connected into Disparity To Z. However, if no camera is connected, the node provides artistic controls for determining a Z channel. The depth created by this node can be used for post effects like fogging or depth of field (DoF).

The Z values produced become more incorrect the larger (negative) they get. The reason is that disparity approaches a constant value as Z approaches -infinity. So Z = -1000 and Z = -10000 and Z = -100000 may map to D=142. 4563 and D=142. 4712 and D=142. 4713. As you can see, there is only 0.0001 in D to distinguish between 10,000 and 100,000 in Z. The maps produced by disparity are not accurate enough to make distinctions like this.

Inputs

The three inputs on the Disparity To Z node are used to connect the left and right images and a camera node.

Left Input: The orange input is used to connect either the left eye image or the stack image.

Right Input: The green input is used to connect the right eye image. This input is available only when the Stack Mode menu is set to Separate.

Stereo Camera: The magenta input is used to connect a stereo camera node.

Outputs

Unlike most nodes in Fusion, Disparity To Z has two outputs for the left and right eye.

Left Output: This holds the left eye image with a new Z channel, or a Stacked Mode image with a new disparity channel.

Right Output: This holds the right eye image with a new Z channel. This output is visible only if Stack Mode is set to Separate.

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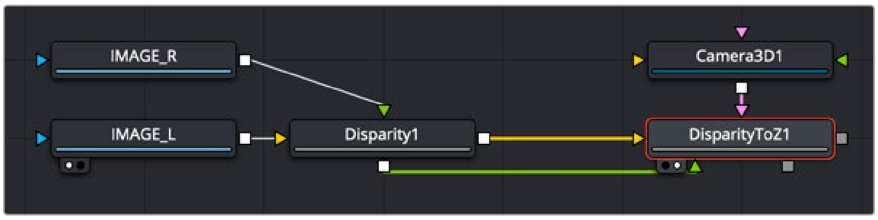
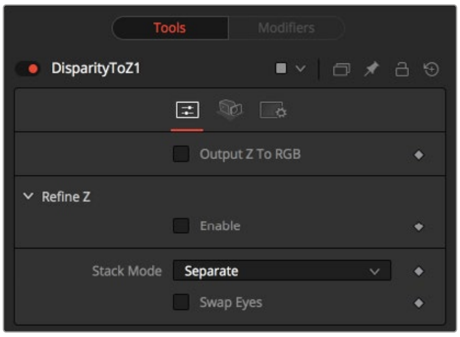
Basic Node Setup

Disparity To Z takes a 3D camera and stereo images containing a disparity channel as inputs. The output is an image with a newly computed Z channel.



A Disparity to Z node creates an image with a Z channel.

Inspector



The Disparity To Z Controls tab

Controls Tab

In addition to outputting Z values in the Z channel, this tab promotes the color channels to float32 and outputs the Z values into the color channels as {z, z, z, 1}. This option is useful to get a quick look at the Z channel.

NOTE: Z values are negative, becoming more negative the further you are from the camera. The viewers only show 0.0 to 1.0 color, so to visualize other data it has to be converted via a normalization method to fit in a display 0-1 range. To do this, right-click in the viewer and choose Options > Show Full Color Range.

Output Z to RGB

Rather than keeping the Z values within the associated aux channel only, they will be copied into the RGB channels for further modification with any of Fusion's nodes.

Refine Z

The Enable checkbox refines the depth map based upon the RGB channels. The refinement causes edges in the flow to align more closely with edges in the color channels. The downside is that unwanted details in the color channels start to show up in the flow. You may want to experiment with using this option to soften out harsh edges for Z-channel post effects like depth of field or fogging.

HiQ Only

Activating this checkbox causes the Refine Z option to process only when rendering is set to High Quality. You can ensure High Quality is enabled by right-clicking to the left or right of the transport controls in the main toolbar.

Strength

Increasing this slider does two things. It smooths out the depth in constant color regions and moves edges in the Z channel to correlate with edges in the RGB channels.

Increasing the refinement has the undesirable effect of causing texture in the color channel to show up in the Z channel. You will want to find a balance between the two.

Radius

This is the radius of the smoothing algorithm.

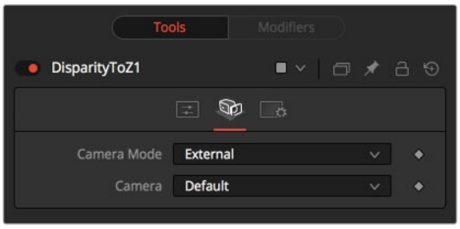
Stack Mode

This menu determines how the input images are stacked.

When set to Separate, the Right Input and Output will appear, and separate left and right images must be connected.

Swap Eyes

Enabling this checkbox causes left and right images to be swapped.



The Disparity To Z Camera tab

Camera Tab

If you need correct real-world Z values because you are trying to match some effect to an existing scene, you should use the External Camera options to get precise Z values back. If any Z-buffer will suffice and you are not that particular about the exact details of how it is offset and scaled, or if there is no camera available, the Artistic option might be helpful.

- **External Mode:** An input is available on the node to connect an existing stereo Camera 3D. This can either be a single stereo Camera 3D (i.e., its eye separation is set to non-zero), or a pair of (tracked) Camera 3Ds connected via the Camera 3D > Stereo > Right Camera input.
- **Artistic Mode:** If you do not have a camera, you can adjust these controls to produce an “artistic” Z channel whose values will not be physically correct but will still be useful. To reconstruct the Disparity > Z Curve, pick (D, Z) values for a point in the foreground and a point in the background.

TIP: If artistic mode is a little too “artistic” for you and you want more physically-based parameters to adjust (e.g., convergence and eye separation), create a dummy Camera 3D, connect it into the Disparity To Z > Camera input, and then fiddle with the Camera 3D’s controls.

Foreground Disparity (Pick from Left Eye)

When the camera Mode is set to Artistic, a Foreground Disparity slider is available. This is the disparity for the closest foreground object. It will get mapped to the depth value specified by the Foreground Depth control. Any objects with disparity outside of the range [ForegroundDisparity, BackgroundDisparity] will have their disparity values clipped to this range leading to flat areas in the Z channel, so make sure that you pick values that enclose the actual disparity range.

Background Disparity (Pick from Left Eye)

When the camera Mode is set to Artistic, a Background Disparity is available. This is the disparity for the furthest background object. It will get mapped to the depth value specified by the Background Depth control. One way to think of this input is as the upper limit to disparity values for objects at -infinity. This value should be for the left eye. The corresponding value in the right eye will be the same in magnitude but negative.

Foreground Depth

This is the depth to which Foreground Disparity will be mapped. Think of this as the depth of the nearest object. Note that values here are positive depth.

Background Depth

This is the depth to which Background Disparity will be mapped. Think of this as the depth of the most distant object.

Falloff

Falloff controls the shape of the depth curve between the requested foreground and background depths. When set to Hyperbolic, the disparity-depth curve behaves roughly like depth = constant/disparity. When set to Linear, the curve behaves like depth = constant * disparity. Hyperbolic tends to emphasize Z features in the foreground, while linear gives foreground/background features in the Z channel equal weighting.

Unless there's a specific reason, choose Hyperbolic, as it is more physically accurate, while Linear does not correspond to nature and is purely for artistic effect.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Global Align [GA]

The GlobalAlign node

NOTE: The Global Align node is available only in Fusion Studio and DaVinci Resolve Studio.

Global Align Node Introduction

As opposed to Stereo Align, this node does not utilize optical flow at all. It's meant as a fast and convenient way to do simple stereo alignment for both X and Y as well as rotation.

Global Align comes in handy at the beginning of the node chain to visually correct major differences between the left and right eye before calculating Disparity.

Manual correction of large discrepancies between left and right, as well as applying an initial color matching, helps Disparity generate more accurate results.

Inputs

The two inputs on the Global Align node are used to connect the left and right images.

Left Input: The orange input is used to connect either the left eye image or the stack image.

Right Input: The green input is used to connect the right eye image. This input is available only when the Stack Mode menu is set to Separate.

Outputs

Unlike most nodes in Fusion, Global Align has two outputs for the left and right eye.

Left Output: This outputs the newly aligned left eye image.

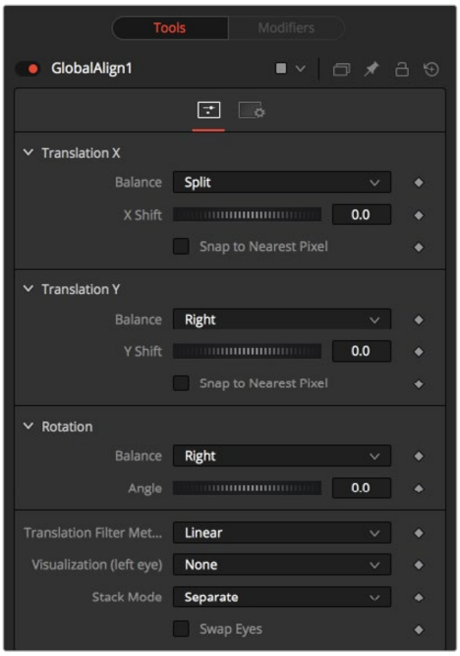
Right Output: This outputs the newly aligned right eye image.

Basic Node Setup

Global Align is typically placed at the beginning of the node tree. Below it is inserted between the left and right eye images and the Disparity node to visually correct major differences.

A Global Align node used to manually correct left and right eye discrepancies

Inspector



The Global Align Controls tab

Controls Tab

The Controls tab includes translation and rotation controls to align the stereo images manually.

Translation X and Y

- **Balance:** Determines how the global offset is applied to the stereo footage.
- **None:** No translation is applied.
- **Left Only:** The left eye is shifted, while the right eye remains unaltered.
- **Right Only:** The right eye is shifted, while the left eye remains unaltered.
- **Split Both:** Left and right eyes are shifted in opposite directions.

Snap to Nearest Pixel

While adjusting the X or Y shift dial, this option ensures that the image is shifted in full pixel amounts only to maintain optimum quality. This avoids sub-pixel rendering of the image, which could result in subtle blurring.

Rotation

- **Balance:** Determines how the global rotation is applied to the stereo footage.
- **None:** No rotation is applied.
- **Left Only:** The left eye is rotated, while the right eye remains unaltered.
- **Right Only:** The right eye is rotated, while the left eye remains unaltered.
- **Split Both:** Left and right eyes are rotated in opposite directions.

Angle

This dial adjusts the angle of the rotation. Keep in mind that the result depends on the Balance settings. If only rotating one eye by, for example, 10 degrees, a full 10-degree rotation will be applied to that eye.

When applying rotation in Split mode, one eye will receive a -5 degree and the other eye a +5 degree rotation.

Translation Filter Method

This menu chooses the filter method that delivers the best results depending on the content of your footage.

Visualization

This control allows for different color encodings of the left and right eye to conveniently examine the results of the above controls without needing to add an extra Anaglyph or Combiner node.

Set this to None for final output.

Stack Mode

Determines how the input images are stacked.

When set to Separate, the right input and output will appear, and separate left and right images must be connected.

Swap Eyes

With Stacked Mode, image stereo pairs' left and right images can be swapped.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

New Eye [NE]

The New Eye node

NOTE: The New Eye node is available only in Fusion Studio and DaVinci Resolve Studio.

New Eye Node Introduction

The New Eye node constructs a new image by interpolating between two existing stereo images using the embedded disparity channels. This node can also be used to replace one view with a warped version of the other. In Stack Mode, L and R outputs will output the same image.

You can map the left eye onto the right eye and replace it. This can be helpful when removing errors from certain areas of the frame.

New Eye does not interpolate the aux channels but instead destroys them. In particular, the disparity channels are consumed/destroyed. Add another Disparity node after the New Eye if you want to generate Disparity for the realigned footage.

Inputs

The two inputs on the New Eye node are used to connect the left and right images.

Left Input: The orange input is used to connect either the left eye image or the stack image.

Right Input: The green input is used to connect the right eye image. This input is available only when the Stack Mode menu is set to Separate.

Outputs

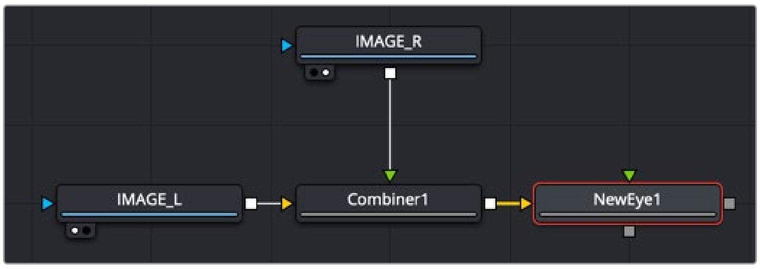
Unlike most nodes in Fusion, New Eye has two outputs for the left and right eye.

Left Output: This outputs the left eye image with a new disparity channel, or a Stacked Mode image with a new disparity channel.

Right Output: This outputs the right eye image with a new disparity channel. This output is visible only if Stack Mode is set to Separate.

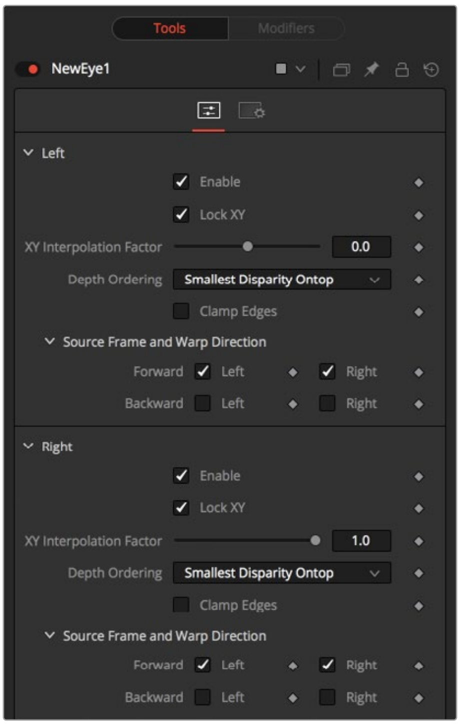
Basic Node Setup

The New Eye node in the example below constructs a new image by interpolating between the two stereo images using the embedded disparity channels.



A New Eye node creates a new stereo image using embedded disparity

Inspector



The New Eye Controls tab

Controls Tab

The Controls tab is divided into identical parameters for the left eye and right eye. The parameters are used to select which eye to recreate and the methods used for the interpolation.

Enable

The Enable checkbox allows you to activate the left or right eye independently. The New Eye will replace enabled eye with an interpolated eye. For example, if the left eye is your “master” eye and you are recreating the right eye, you would disable the left eye and enable the right eye.

Lock XY

Locks the X and Y interpolation parameters. When they are unlocked, you can provide separate interpolation factors for using the X and Y disparity. For example, if you are working with the right eye and you have the X Interpolation slider set to 1.0 and the Y Interpolation slider set to -1.0, you will be effectively interpolating the left eye onto the right eye but vertically aligned to the left eye.

XY Interpolation Factor

Interpolation determines where the interpolated frame is positioned, relative to the two source frames: A slider position of -1.0 outputs the frame Left and a slider position of 1.0 outputs the frame Right. A slider position of 0.0 outputs a result that is halfway between Left and Right.

Depth Ordering

The Depth Ordering is used to determine which parts of the image should be rendered on top. When warping images, there is often overlap. When the image overlaps itself, there are two options for which values should be drawn on top.

- **Largest Disparity On Top:** The larger disparity values will be drawn on top in the overlapping image sections.
- **Smallest Disparity On Top:** The smaller disparity values will be drawn on top in the overlapping image sections.

Clamp Edges

Under certain circumstances, this option can remove the transparent gaps that may appear on the edges of interpolated frames. Clamp Edges will cause a stretching artifact near the edges of the frame that is especially visible with objects moving through it or when the camera is moving.

Because of these artifacts, it is a good idea to use Clamp Edges only to correct small gaps around the edges of an interpolated frame.

Softness

Helps to reduce the stretchy artifacts that might be introduced by Clamp Edges.

If you have more than one of the Source Frame and Warp Direction checkboxes turned on, this can lead to doubling up of the stretching effect near the edges. In this case, you'll want to keep the softness rather small at around 0.01. If you have only one checkbox enabled, you can use a larger softness at around 0.03.

Source Frame and Warp Direction

The output of this node is generated by combining up to four different warps. You can choose to use either the color values from the left or right frame in combination with the Forward (left > right) Disparity or the Backward (right > left) Disparity. Sometimes you will want to replace an existing eye. For example, if you want to regenerate the right eye, you would only use left eye warps.

It's good to experiment with various options to see which gives the best effect. Using both the left and right eyes can help fill in gaps on the left/right side of images. Using both the Forward/Backward Disparity can give a doubling-up effect in places where the disparities disagree with each other.

- **Left Forward:** Takes the Left frame and uses the Forward Disparity to interpolate the new frame.
- **Right Forward:** Takes the Right frame and uses the Forward Disparity to interpolate the new frame.
- **Left Backward:** Takes the Left frame and uses the Back Disparity to interpolate the new frame.
- **Right Backward:** Takes the Right frame and uses the Back Disparity to interpolate the new frame.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Splitter [Spl]

The Splitter node

NOTE: The Splitter node is available only in Fusion Studio and DaVinci Resolve Studio.

Splitter Node Introduction

The Splitter takes a stacked input image—for example, created with the Combiner—and provides two output images: a left eye and a right eye.

Inputs

The two inputs on the Splitter node are used to connect the left and right images.

Left Input: The orange input is used to connect a stacked stereo image.

Outputs

Unlike most nodes in Fusion, the Splitter node has two outputs for the left and right eye.

Left Output: This outputs the left eye image.

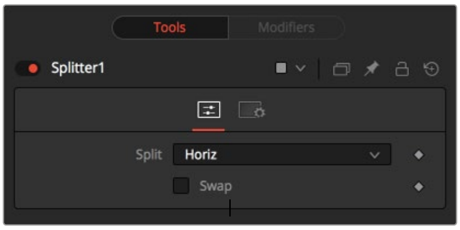
Right Output: This outputs the right eye image.

Basic Node Setup

Below, a stacked stereo image is connected to the input on a Splitter where a left and right eye is output.

A Splitter node creates a left and right image from a stacked stereo image

Inspector



The Splitter Controls tab

Controls Tab

The Controls tab is used to define the type of stacked image connected to the node's input.

Split

The Split menu contains three options for determining the orientation of the stacked input image.

- **None:** No operation takes place. The output image on both outputs is identical to the input image.
- **Horiz:** The node expects a horizontally stacked image. This will result in two output images, each being half the width of the input image.
- **Vert:** The node expects a vertically stacked image. This will result in two output images, each being half the height of the input image.

Swap Eyes

Allows you to easily swap the left and right eye outputs.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Stereo Align [SA]

The StereoAlign node

NOTE: The Stereo Align node is available only in Fusion Studio and DaVinci Resolve Studio.

Stereo Align Node Introduction

This extremely versatile node for fixing Stereo issues can be used for performing any of the following actions or combinations thereof:

- Vertical alignment of one eye to the other
- Changing the convergence
- Changing the eye separation

By combining these operations in one node, you can execute them using only a single image resampling. In essence, this node can be thought of as applying scales and translation to the disparities and then using the modified disparities to interpolate between the views.

NOTE: Changing the eye separation can cause holes to appear, and it may not be possible to fill them since the information needed may not be in either image. Even if the information is there, the disparity may have mismatched the holes. You may need to fill the holes manually. This node modifies only the RGBA channels.

TIP: Stereo Align does not interpolate the aux channels but instead destroys them. In particular, the disparity channels are consumed/destroyed. Add another Disparity node after the StereoAlign if you want to generate Disparity for the realigned footage.

Inputs

The two inputs on the Stereo Align node are used to connect the left and right images.

Left Input: The orange input is used to connect either the left eye image or the stack image.

Right Input: The green input is used to connect the right eye image. This input is available only when the Stack Mode menu is set to Separate.

Outputs

Unlike most nodes in Fusion, Stereo Align has two outputs for the left and right eye.

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Left Output: This outputs the left eye image with a new disparity channel, or a Stacked Mode image with a new disparity channel.

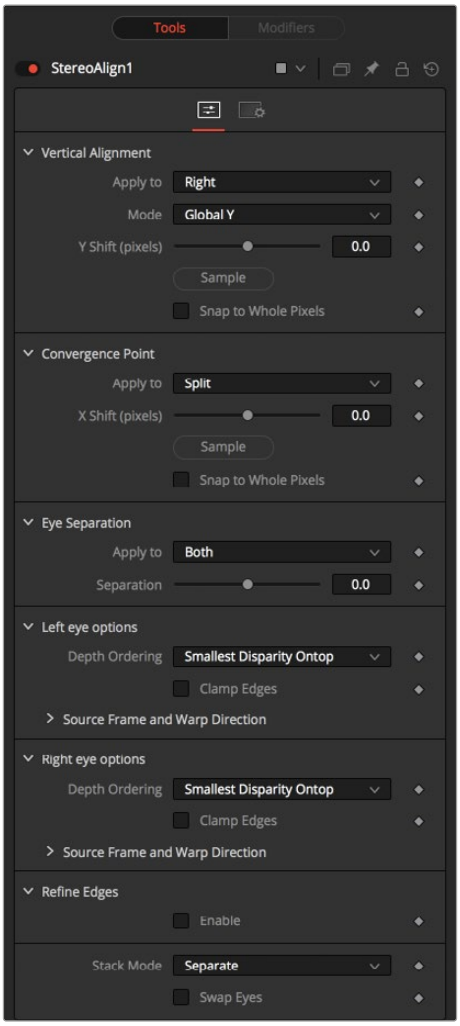
Right Output: This outputs the right eye image with a new disparity channel. This output is visible only if Stack Mode is set to Separate.

Basic Node Setup

Below, the Stereo Align receives the left and right eye images with disparity. Once the adjustments are made, another Disparity node is added after it to generate disparity for the realigned footage.

A Stereo Align node destroys the disparity channel, so another Disparity node is placed after it

Inspector



The Stereo Align Controls tab

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Controls Tab

Vertical Alignment

This option determines how the vertical alignment is split between two eyes. Usually, the left eye is declared inviolate, and the right eye is aligned to it to avoid resampling artifacts.

When doing per pixel vertical alignment, it may be helpful to roughly pre-align the images by a global Y-shift before disparity computation because the disparity generation algorithm can have problems resolving small objects that move large distances.

Also, be aware that you must be careful about lens distortion because even if two cameras are perfectly vertically aligned, they will still have vertical disparities due to lens distortion. As a best practice, remove the lens distortion before computing the disparity. When a vertical alignment of the right eye is done, you have essentially removed the Y-component of the lens distortion in the right eye, and it will look wrong later when you try to distort it again.

Apply to

- **Right:** Only the right eye is adjusted.
- **Left:** Only the left eye is adjusted.
- **Both:** The vertical alignment is split evenly between the left and right eyes.

Mode

- **Global:** The eyes are simply translated up or down by the Y-shift to match up.
- **Per Pixel:** The eyes are warped pixel-by-pixel using the disparity to vertically align.

> Keep in mind that this can introduce sampling artifacts and edge artifacts.

Y-shift

Y-shift is available only when the Mode menu is set to Global. You can either adjust the Y-shift manually to get a match or drag the Sample button into the viewer, which picks from the disparity channel of the left eye. Also remember, if you use this node to modify disparity, you can't use the Sample button while viewing the node's output.

Snap to Whole Pixels

You can snap the global shift to whole pixels by enabling this option. When enabled, there is no resampling of the image, but rather a simple shift is done so there will be no softening or image degradation.

Convergence Point

The Convergence Point section is used as a global X-translation of L/R images.

Apply to

This menu determines which eyes are affected by convergence. You can choose to apply the convergence to the left eye, right eye, or split between the two. In most cases, this will be set to Split. If you set the eyes to Split, then the convergence will be shared 50-50 between both eyes. Sharing the convergence between both eyes means you get half the shift in each eye, which in turn means smaller holes and artifacts that need to be fixed later. The tradeoff is that you've resampled both eyes rather than keeping one eye as a pure reference master.

X-shift

You can either use the slider to adjust the X-shift manually to get a match or use the Sample button to pick from the disparity channels for easy point-to-feature alignment.

Snap

You can snap the global shift to whole pixels using this option. In this mode, there is no resampling of the image, but rather a simple shift is done so there will be no softening or image degradation.

Eye Separation

Eye Separation changes the distance between the left/right eyes, causing objects in the left/right eyes to converge/diverge further depending on their distance from the camera.

This has the same effect as the Eye Separation option in the Camera 3D node.

Separation

This is a scale factor for eye separation.

- When set to 0.0, this leaves the eyes unchanged.
- Setting it to 0.1 increases the shifts of all objects in the scene by a factor of 10% in each eye.
- Setting it to 0.1 will scale the shifts of all objects 10% smaller.

Unlike the Split option for vertical alignment, which splits the alignment effect 50-50 between both eyes, the Both option will apply 100-100 eye separation to both eyes. If you are changing eye separation, it can be a good idea to enable per-pixel vertical alignment, or the results of interpolating from both frames can double up.

Left/Right Eye Options

The left and right eye options contain depth ordering and warp direction controls independently for the left and right eye.

Depth Ordering

The Depth Ordering is used to determine which parts of the image should be rendered on top. When warping images, there is often overlap. When the image overlaps itself, there are two options for which values should be drawn on top.

- **Largest Disparity On Top:** The larger disparity values will be drawn on top in the overlapping image sections.
- **Smallest Disparity On Top:** The smaller disparity values will be drawn on top in the overlapping image sections.

Clamp Edges

Under certain circumstances, this option can remove the transparent gaps that may appear on the edges of interpolated frames. Clamp Edges will cause a stretching artifact near the edges of the frame that is especially visible with objects moving through it or when the camera is moving.

Because of these artifacts, it is a good idea to use Clamp Edges only to correct small gaps around the edges of an interpolated frame.

Edge Softness

Helps to reduce the stretchy artifacts that might be introduced by Clamp Edges.

If you have more than one of the Source Frame and Warp Direction checkboxes turned on, this can lead to doubling up of the stretching effect near the edges. In this case, you'll want to keep the softness rather small at around 0.01. If you have only one checkbox enabled, you can use a larger softness at around 0.03.

Source Frame and Warp Direction

The output of this node is generated by combining up to four different warps. You can choose to use either the color values from the left or right frame in combination with the Forward (left > right)

Disparity or the Backward (right > left) Disparity. Sometimes you will want to replace an existing eye. For example, if you want to regenerate the right eye, you would use only left eye warps.

It's good to experiment with various options to see which gives the best effect. Using both the left and right eyes can help fill in gaps on the left/right side of images. Using both the Forward/Backward Disparity can give a doubling-up effect in places where the disparities disagree with each other.

- **Left Forward:** Takes the Left frame and uses the Forward Disparity to interpolate the new frame.
- **Right Forward:** Takes the Right frame and uses the Forward Disparity to interpolate the new frame.
- **Left Backward:** Takes the Left frame and uses the Back Disparity to interpolate the new frame.
- **Right Backward:** Takes the Right frame and uses the Back Disparity to interpolate the new frame.

Stack Mode

In Stack Mode, L and R outputs will output the same image.

If High Quality is off, the interpolations are done using nearest-neighbor sampling, leading to a more "noisy" result. To ensure High Quality is enabled, right-click under the viewers, near the transport controls, and choose High Quality from the pop-up menu.

Swap Eyes

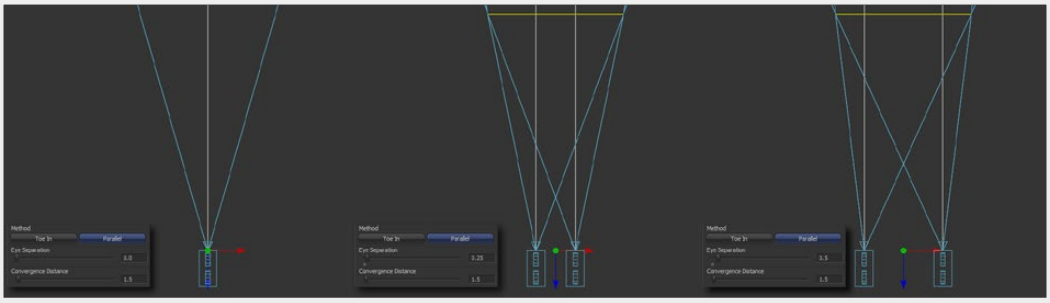
Allows you to easily swap the left and right eye outputs.

Common Controls

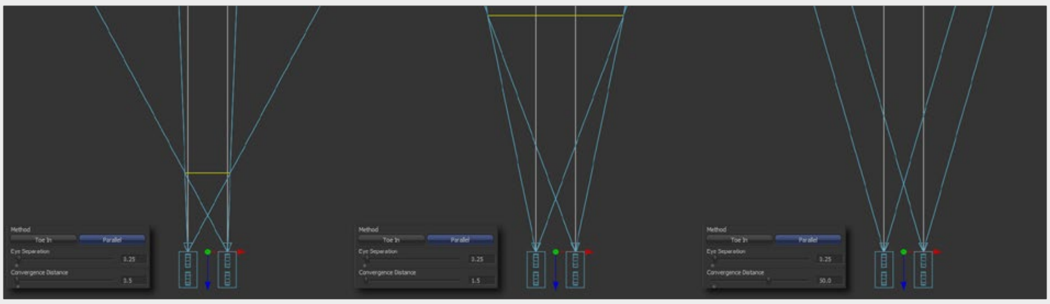
Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Example



Different settings for Eye Separation...



...and example settings for Convergence

Z To Disparity [Z2D]

The Z To Disparity node

NOTE: The Z To Disparity node is available only in Fusion Studio and DaVinci Resolve Studio.

Z To Disparity Node Introduction

Z To Disparity takes a stereo camera and an image containing a Z channel and outputs the same image but with disparity channels in it. This is useful for constructing a Disparity map from CG renders, which will be more accurate than the Disparity map created from the Disparity node.

Inputs

The three inputs on the Z To Disparity node are used to connect the left and right images and a camera node.

Left Input: The orange input is used to connect either the left eye image or the stack image.

Right Input: The green input is used to connect the right eye image. This input is available only when the Stack Mode menu is set to Separate.

Stereo Camera: The magenta input is used to connect a stereo perspective camera, which may be either a Camera 3D with eye separation, or a tracked L/R Camera 3D.

Outputs

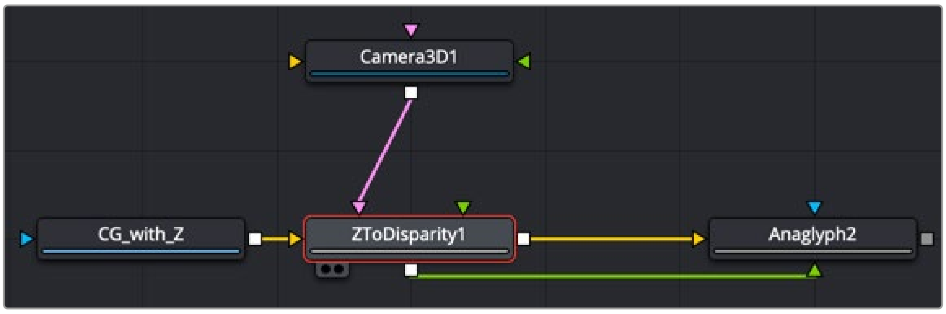
Unlike most nodes in Fusion, Z To Disparity has two outputs for the left and right eye.

Left Output: This outputs the left eye image containing a new disparity channel, or a Stacked Mode image with a new disparity channel.

Right Output: This outputs the right eye image with a new disparity channel. This output is visible only if Stack Mode is set to Separate.

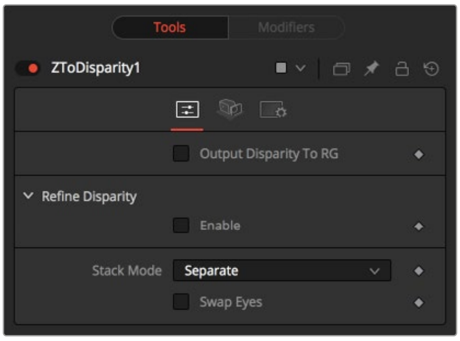
Basic Node Setup

Below, a stereo camera and an image containing a Z channel is connected to a Z To Disparity node. The same image is output with disparity channels.



A Z To Disparity node takes an image with a Z channel and creates a disparity channel

Inspector



The Z To Disparity Controls tab

Controls Tab

The Controls tab includes settings that refine the conversion algorithm.

Output Disparity To RGB

In addition to outputting disparity values into the disparity channel, activating this checkbox causes Z To Disparity to also output the disparity values into the color channels as {x, y, 0, 1}.

When activated, this option will automatically promote the RGBA color channels to float32. This option is useful for a quick look to see what the disparity channel looks like.

Refine Disparity

This refines the Disparity map based on the RGB channels.

Strength

Increasing this slider does two things. It smooths out the depth in constant color regions and moves edges in the Z channel to correlate with edges in the RGB channels. Increasing the refinement has the undesirable effect of causing texture in the color channel to show up in the Z channel. You will want to find a balance between the two.

Radius

This is the pixel-radius of the smoothing algorithm.

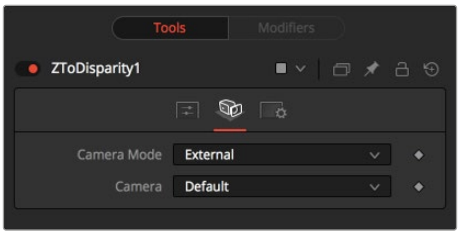
Stack Mode

In Stack Mode, L and R outputs will output the same image.

If HiQ is off, the interpolations are done using nearest-neighbor sampling, leading to a more “noisy” result.

Swap Eyes

This allows you to easily swap the left and right eye outputs.



The Z To Disparity Camera tab

Camera Tab

The Camera tab includes settings for selecting a camera and setting its conversion point if necessary.

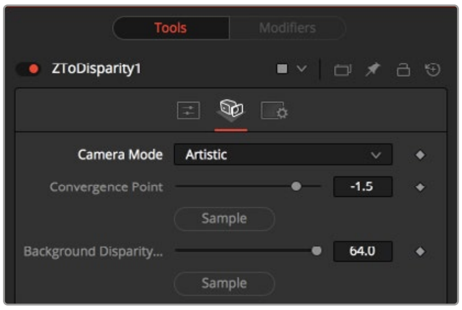
Camera Mode

If you need correct real-world disparity values because you are trying to match some effect to an existing scene, you should use the External setting to get precise disparity values back. When External is selected, a magenta camera input is available on the node to connect an existing stereo Camera 3D node, and use the Camera settings to determine the Disparity settings.

If you just want any disparity and do not particularly care about the exact details of how it is offset and scaled, or if there is no camera available, then the Artistic setting might be helpful.

Camera

If you connect a Merge 3D node that contains multiple cameras to the camera input, the Camera menu allows you to select the camera to use.



Artistic Camera mode

If you do not have a camera, you can adjust the artistic controls to produce a custom disparity channel whose values will not be physically correct but will be good enough for compositing hacks. There are two controls to adjust:

Convergence Point

This is the Z value of the convergence plane. This corresponds to the negative of the Convergence Distance control that appears in Camera 3D. At this distance, objects in the left and right eyes are at exactly the same position (i.e., have zero disparity).

Objects that are closer appear to pop out of the screen, and objects that are further appear behind the screen.

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Background Disparity (Sample from Left Eye)

This is the disparity of objects in the distant background. You can think of this as the upper limit to disparity values for objects at infinity. This value should be for the left eye. The corresponding value in the right eye will be the same in magnitude but negative.

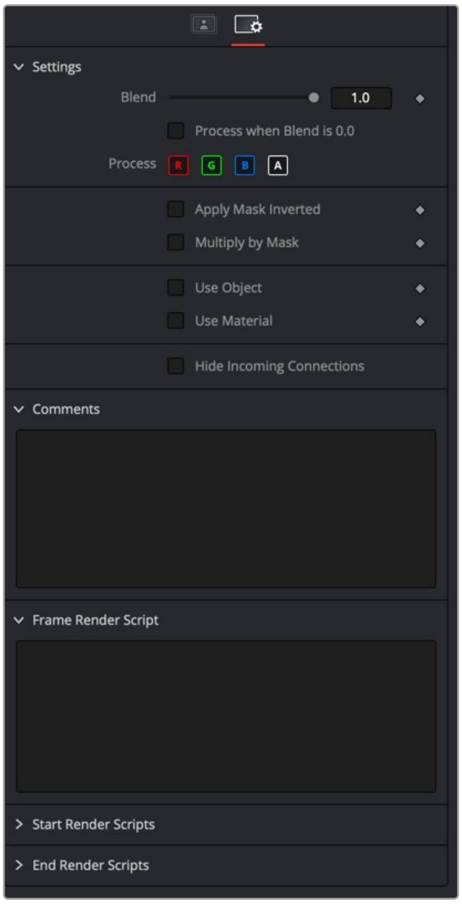
Settings Tab

The Settings tab in the Inspector is also duplicated in other Stereo nodes. These common controls are described in detail in the following "The Common Controls" section.

The Common Controls

The Common Settings tab can be found on virtually every tool found in Fusion. The following controls are specific settings for Stereo nodes.

Settings Tab



The Common Settings Stereo 3D Settings tab

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this will cause the tool to skip processing entirely, copying the input straight to the output.

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Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if the node is scripted to trigger a task, but the node's value is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur will first be applied to the image, and then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this will generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the mask channel's values. This will cause all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels will be used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

> For more information on the Coverage and Background Color channels, *see Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID will be used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the field will hide that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line will reappear.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.